THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS (PART 1 + PART 2)

Chapters 1–14 | Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 04	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (−6, −2) lies in the: **1**

(a) fourth quadrant (b) first quadrant
(c) third quadrant (d) second quadrant

2. The point on the y-axis which is 5 units below the origin is: **1**

(a) (0, −5) (b) (5, 0) (c) (0, 5) (d) (−5, 0)

3. The solution of the pair of equations x + y = 5 and x − y = 1 is: **1**

(a) x = 2, y = 3 (b) x = 4, y = 1 (c) x = 3, y = 2 (d) x = 5, y = 0

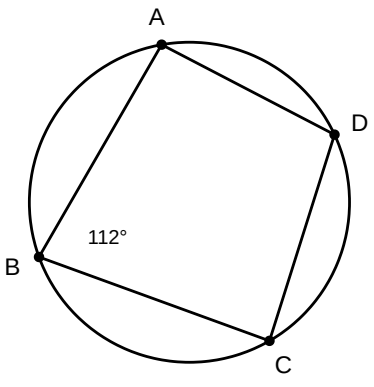
4. Which of the following is an irrational number? **1**

(a) 0.1010010001… (the number of zeros keeps increasing)
(b) 0.25
(c) 3.14
(d) 0.101010… (10 repeating)

5. The product $(x - 3)(x^2 + 3x + 9)$ is equal to: **1**

(a) $x^3 - 9$ (b) $x^3 + 27$ (c) $(x - 3)^3$ (d) $x^3 - 27$

6. ABCD is a cyclic quadrilateral with ∠ABC = 112°. Then ∠ADC is: **1**



(a) 112° (b) 98° (c) 78° (d) 68°

7. The perpendicular from the centre of a circle to a chord is 12 cm long and the chord is 18 cm long. The radius of the circle is: **1**

(a) 30 cm (b) $\sqrt{63}$ cm (c) 15 cm (d) 21 cm

8. An isosceles triangle has equal sides of 17 cm and base 16 cm. Its area, by Heron's formula, is: **1**

(a) 120 cm² (b) 136 cm² (c) 240 cm² (d) 60 cm²

9. A card is drawn at random from cards numbered 1 to 20. The probability that it is a prime number is: **1**

(a) $\frac{2}{5}$ (b) $\frac{9}{20}$ (c) $\frac{1}{2}$ (d) $\frac{3}{10}$

10. If $t_n = (-1)^n(3n - 1)$, then t_5 is: **1**

(a) 14 (b) −14 (c) 16 (d) −16

11. The converse of 'If two lines are parallel, then the corresponding angles formed by a transversal are equal' is: **1**

(a) If the corresponding angles are not equal, the lines are parallel.

(b) If the corresponding angles formed by a transversal are equal, then the two lines are parallel.

(c) If two lines are not parallel, then the corresponding angles are not equal.

(d) Parallel lines have equal alternate angles.

12. A counterexample to 'If a number ends in 5, then it is a multiple of 10' is: **1**

(a) 10 (b) 30 (c) 20 (d) 15

13. A class has 12 boys of average height 150 cm and 18 girls of average height 145 cm. The average height of the class is: **1**

(a) 148 cm (b) 147.5 cm (c) 146 cm (d) 147 cm

14. A cone has base radius 5 cm and slant height 13 cm. Its height is: **1**

(a) 18 cm (b) 8 cm (c) 12 cm (d) $\sqrt{194}$ cm

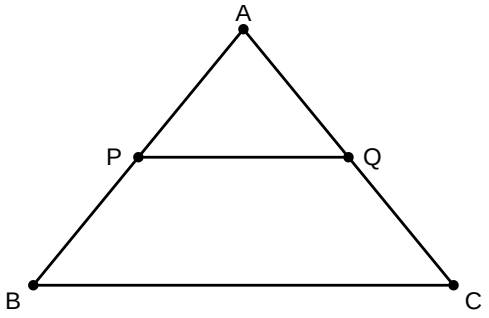
15. If the last step (writing down the final carry) of the addition algorithm is left out, the result for 85 + 47 would be: **1**

(a) 132 (b) 42 (c) 32 (d) 122

16. In a parallelogram ABCD, $\angle A : \angle B = 4 : 5$. Then $\angle A$ is: **1**

(a) 90° (b) 72° (c) 80° (d) 100°

17. In $\triangle ABC$, P is the midpoint of AB and PQ ∥ BC meets AC at Q. If AC = 14 cm, then CQ is: **1**



(a) 10 cm (b) 14 cm (c) 3.5 cm (d) 7 cm

18. Which term of the AP 7, 11, 15, … is 63? **1**

(a) 15th (b) 14th (c) 16th (d) 13th

19. **Assertion (A):** The point (−3, 0) lies on the y-axis. **1**
Reason (R): A point on the y-axis has its abscissa equal to zero.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The simple average of two group averages equals the combined average only when the two groups are of equal size. **1**
Reason (R): The combined average is the total of all the values divided by the total number of values.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

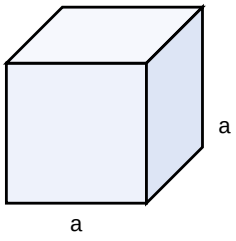
21. A library charges a fine of ₹10 for the first 3 days a book is overdue and ₹4 for every extra day. Find the fine for a book that is 10 days overdue. **2**

22. Without actual division, state which of $\frac{7}{24}$ and $\frac{7}{32}$ has a terminating decimal expansion. Give reasons. **2**

OR

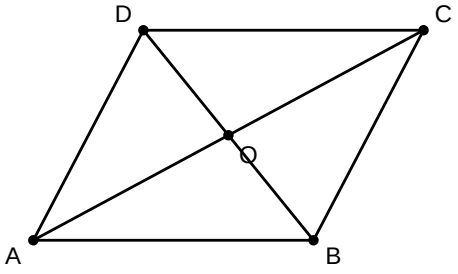
Find two rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.

23. The total surface area of a cube is 150 cm². Find the length of its edge and its volume. **2**



24. Write the converse of ‘If a number is a multiple of 10, then it is a multiple of 5.’ Is the converse true? Justify. **2**

25. In the parallelogram ABCD, the diagonals meet at O. If OA = (3x − 2) cm and OC = (x + 6) cm, find x and the length of AC. **2**



SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that 0.2777… (7 repeating) = $\frac{5}{18}$. **3**

27. In a survey of 60 students, the pets they keep were recorded: **3**

Pet	Dog	Cat	Fish	None
Students	20	15	10	15

A student is chosen at random. Find the probability that the student (i) keeps a dog, (ii) keeps some pet, (iii) keeps a cat or a fish.

OR

A die was thrown 100 times and the outcomes recorded:

Outcome	1	2	3	4	5	6
Frequency	14	18	16	20	17	15

Find the experimental probability of getting (i) an even number, (ii) a number greater than 4, (iii) a number less than 3.

28. Draw the graphs of the equations $3x + y = 9$ and $x + y = 5$ on the same grid and use them to find the solution of the pair. **3**

29. A white stork flew 40 km per day for 3 days and 55 km per day for 2 days. Find its average daily distance over the 5 days and explain why (40 + 55)/2 would be wrong. **3**

30. In a circle with centre O, $\angle AOB = 100°$. Find $\angle ACB$, where C is a point on the major arc, and $\angle ADB$, where D is a point on the minor arc. **3**

OR

Using Heron's formula, find the area of a triangle with sides 25 cm, 52 cm and 63 cm.

31. Prove that the diagonals of a parallelogram bisect each other. **3**

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. The corners of a square flower bed are marked on a grid as A(1, 0), B(4, 1), C(3, 4) and D(0, 3). **5**
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths of the four sides. (1½ marks)
(c) Find the lengths of the diagonals AC and BD and justify that ABCD is a square. (1½ marks)

33. The seats in a hall are arranged in 15 rows. The first row has 18 seats and every next row has 2 seats more than the row before. Find the number of seats in the last row and the total number of seats. (2 + 3) **5**

OR

A company tested 500 batteries and recorded their lifetimes:

Life (hours)	0 – 100	100 – 200	200 – 300	300 – 400
Number of batteries	30	90	210	170

A battery is chosen at random. Find the probability that it lasts (i) at least 200 hours (1), (ii) less than 100 hours (1), (iii) between 100 and 300 hours (2); (iv) verify that the four class probabilities add up to 1 (1).

34. Prove the converse of the Midpoint Theorem: the line drawn through the midpoint of one side of a triangle, parallel to another side, bisects the third side. **5**

OR

A chemist has 200 mL of a 15% acid solution. How many millilitres of a 40% acid solution must be added to obtain a 20% solution?

35. The product of two consecutive positive odd integers is 143. Form a quadratic expression in the smaller integer x, factorise it and find the integers. **5**

OR

A rectangle is formed from 3 square tiles of side x units, 10 strips of size x by 1 unit and 8 unit squares. Write the total area as a polynomial, factorise it and give the dimensions of the rectangle.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A shopkeeper stacks tins in a display. The bottom row has 20 tins and every row above it has 2 tins fewer than the row below. The number of tins in the n-th row from the bottom is $t_n = 22 - 2n$.

(i) How many tins are in the first row? 1

(ii) How many tins are in the 10th row? 1

(iii) Find the total number of tins in the display if it has 10 rows. 2

OR

(iii) Which row has exactly 8 tins?

37. A triangular park has sides 15 m, 41 m and 52 m.

(i) Find the semi-perimeter of the park. 1

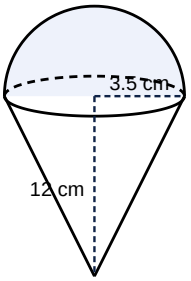
(ii) Find the area of the park using Heron's formula. 1

(iii) Find the length of the perpendicular from the opposite vertex to the longest side. 2

OR

(iii) Grass is laid on the whole park at ₹15 per m². Find the total cost.

38. An ice-cream seller serves ice cream in a cone of base radius 3.5 cm and height 12 cm, with a hemispherical scoop of the same radius on top (π = 22/7).



(i) Find the volume of ice cream inside the cone. 1

(ii) Find the curved surface area of the hemispherical top. 1

(iii) Find the slant height of the cone and its curved surface area. 2

OR

(iii) Find the total outer surface area of the cone and the hemisphere together (excluding the base).